



আন্তর্জাতিক ইসলামি বিশ্ববিদ্যালয় চট্টগ্রাম
الجامعة الإسلامية العالمية شيتاغونغ
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PROJECT REPORT

PROJECT TITLE : Flutter Portfolio Application and SD2
Lab Learning

COURSE CODE : CSE-3644

COURSE TITLE : Software Development II Lab

SUBMITTED TO:

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1. Introduction

The Software Development 2 (SD2) Lab course introduced various concepts of modern software development, including application architecture, state management, responsive user interface design, reusable components, animations, and version control. To apply these concepts practically, I developed a **Professional Flutter Portfolio Application** using Flutter and GetX Architecture.

The application presents my personal information, education, technical skills, projects, and contact details through a modern and responsive interface. This report explains how the project reflects the concepts and practical knowledge gained during the SD2 Lab course.

2. Reflection on SD2 Lab Learning Outcomes

2.1 Flutter Application Development

One of the main objectives of the SD2 Lab course was to learn modern application development using Flutter. During this project, I learned how to build a complete application using Flutter widgets, layouts, and Material 3 design principles. I also learned how to organize a project properly so that it becomes easier to maintain and extend in the future.

2.2 GetX Architecture

The project follows **GetX Architecture**, which separates the application into different layers such as **Views, Controllers, Models, Bindings, and Routes**. This architecture keeps the project clean and organized by separating the user interface from the business logic.

Using GetX also made dependency injection and route management much easier. Each controller is created only when needed, which improves application performance and code maintainability.

2.3 State Management

State management was one of the most important topics covered in SD2 Lab. In this project, I used GetX reactive state management to manage application data.

Whenever the application state changes, the user interface updates automatically without rebuilding the entire screen. This approach improves performance, reduces unnecessary code, and makes the application easier to manage.

2.4 Responsive User Interface

Another important lesson from the course was designing responsive applications that work on different screen sizes.

This project supports desktop, tablet, and mobile devices. The desktop version uses a horizontal navigation bar and wider layouts, while the mobile version automatically changes to a drawer menu and single-column layout. This ensures a consistent and user-friendly experience across different devices.

2.5 Reusable Widgets

The SD2 Lab course emphasized writing reusable and maintainable code. Instead of creating the same UI multiple times, I developed reusable widgets such as:

- Navigation Button
- Section Header
- Custom Button
- Animated Profile Image
- Animated Skill Card
- Animated Project Card
- Footer

Using reusable widgets reduced code duplication and made future modifications much easier.

2.6 UI Animations

To improve the user experience, I implemented several animations throughout the application.

The profile image includes size and color transition effects. Skill cards and project cards animate smoothly when hovered over, while navigation buttons provide animated visual feedback. These animations make the application more interactive and demonstrate the animation concepts learned during the course.

2.7 Project Organization

The project is organized using a modular folder structure.

Different folders are used for:

- Bindings
- Controllers
- Models
- Routes
- Theme
- Views
- Widgets

This structure makes the application easier to understand, debug, and expand.

2.8 Git and GitHub

Throughout the project development process, I used Git for version control and GitHub to store and manage the project repository.

This helped me understand how software projects are managed, how changes are tracked, and how projects can be shared with others using version control systems.

3. Skills Gained

By completing this project, I gained practical experience in many areas of software development.

Some of the important skills I developed include:

- Flutter application development
- GetX Architecture
- State management
- Dependency injection
- Routing
- Responsive UI design
- Material 3 design
- Flutter animations
- Reusable widget development
- Clean code organization
- Git and GitHub
- Problem-solving and debugging

4. Challenges Faced

During the development of this project, I faced several challenges.

Initially, organizing the project according to GetX Architecture was difficult because different files had to be separated into controllers, views, bindings, and models. Making the application responsive for different screen sizes also required careful layout planning.

Implementing smooth animations without affecting application performance was another challenge. Through research and practice, I successfully solved these problems and improved both my coding and debugging skills.

5. Technologies Used

The following technologies and tools were used to develop the project:

- **Flutter** – Used as the main framework for developing the cross-platform mobile application.

- **Dart** – The programming language used for Flutter application development.
- **GetX** – Used for state management, dependency injection, and route management.
- **Flutter Animation Framework** – Used to implement size and color transition animations.
- **Visual Studio Code / Android Studio** – Used as the development environment.
- **Git & GitHub (Optional)** – Used for version control and project management.

6. Conclusion

The **Professional Flutter Portfolio Application** successfully demonstrates the practical implementation of the concepts learned in the **Software Development 2 (SD2) Lab** course.

This project reflects my understanding of Flutter development, GetX Architecture, state management, responsive UI design, reusable widgets, animations, and version control. It also helped me improve my programming, problem-solving, and software design skills.

Overall, this project strengthened my confidence in developing modern Flutter applications and provided valuable practical experience that will help me in future academic and professional software development projects.